3(a)	acceleration (directly) proportional to displacement	**B1**
	acceleration is in opposite direction to displacement	**B1**
3(b)	$\omega^2 = 2k / m$ **and** $\omega = 2\pi f$	**C1**
	$(2\pi f)^2 = (2 \times 130) / 0.84$	**C1**
	$f = 2.8$ Hz	**A1**
3(c)(i)	resonance	**B1**
3(c)(ii)	oscillator supplies energy (continuously)	**B1**
	energy of trolley constant so energy must be dissipated **or** without loss of energy the amplitude would continuously increase	**B1**

Question	Answer	Marks
4	(ultrasound) pulse	**B1**
	reflected at boundaries	**B1**
	gel is used to minimise reflection at skin **or** generated and detected by quartz crystal	**B1**
	time delay between generation and detection gives information about depth	**B1**
	intensity (of reflected wave) gives information about nature of boundary	**B1**